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| **Fiscal Year:** | FY 2016 | **Task Last Updated:** | FY 01/14/2016 |
| **PI Name:** | Motil, Brian J. Ph.D. | | |
| **Project Title:** | Modeling and Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Fluid physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 44135 | **Congressional District:** | 10 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities |
| **Start Date:** | 03/01/2014 | **End Date:** | 05/31/2017 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Motil, Brian J. | **Contact Phone:** | 216-433-6617 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: End date changed to 5/31/2017 per NSSC information (Ed., 4/28/21)<br>NOTE: Extended to 2/28/2018 per PI (Ed., 1/17/17) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Balakotaiah , Vemuri ( University of Houston (grant NNX14AD28G) ) | | |
| **Grant/Contract No.:** | Internal Project ; CoI grant NNX14AD28G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | NOTE: Continuation of "Experimental Studies on Gas-Liquid Two-Phase Flow Through Packed Beds in Microgravity," internal project--Principal Investigator Dr. Brian Motil ; CoInvestigator Vemuri Balakotaiah, Ph.D., grant NNX10AL37G. Period of performance was 4/12/2010-10/11/2013.<br>A basic understanding of two-phase flows through porous media is of interest in many chemical and biological processing systems as well as many geophysical applications. NASA technologies include adiabatic two-phase flows or two-phase flows with heat transfer that are relevant to life support systems, fuel cells, in-situ resource utilization, heat pipes, materials processing, production of pharmaceutical grade water, and transport of nutrients in soils. A very important unit operation that involves gas-liquid flow through porous media is the fixed Packed Bed Reactor (PBR). Examples of current space systems in which a fixed PBR is used include the Volatile Removal Assembly (VRA), the Integrated Advanced Water Recovery System (AWRS), and IntraVenous Water GENeration system (IVGEN). NASA is currently operating systems with packed bed reactors for ground and flight-based testing with an incomplete understanding of how the reduced gravity environment affects the performance and reliability of these reactors. This is especially critical when gas-liquid flows are involved. The expected outcome of this research effort is to develop a set of guidelines and tools to enable engineers to reliably design and operate packed bed reactors for microgravity as well as the lunar and Martian environments.<br><br>Existing approaches for predicting the flow regimes and pressure drop for gas-liquid flow through a PBR are valid only for 1 g, and are based on liquid (or gas) holdup within the bed. Initial microgravity tests conducted in the reduced gravity aircraft and drop tower clearly show that the liquid holdup is significantly influenced by gravity. Furthermore, most of these models have only been verified using the "trickle flow" regime which does not exist in microgravity. Basic hydrodynamic models developed specifically for reduced gravity have been partially validated within the constraints of aircraft and ground facilities. However, typical flow ranges used for real systems require up to several minutes to develop fully which is far beyond the time scales available for ground tests. In addition, instabilities and hysteresis effects that may exist require long testing durations to establish new operating conditions. Thus, testing on the International Space Station (ISS) is required for complete validation of the semi-empirical models.<br><br>Flight experiments can provide critical hydrodynamic information for a project with a broader scope which includes reduced gravity aircraft and ground-based (1 g) experiments. The main objective of the overall project is to develop and validate macroscopic equations that can be used in partial and microgravity conditions to accurately predict flow pattern transitions; pressure drops; and mass transfer rates in chemical and biological reactors in which gas-liquid flows occur through randomly packed beds. The hydrodynamic investigations addressed in the Packed Bed Reactor Experiment (PBRE) focus on the transitions between flow regimes (e.g., bubbly-to-pulse flow transition) and the associated pressure gradients over the range of relevant operating regimes of the PBRs (e.g., liquid and gas flow rates, particle sizes, and fluid viscosities). These design tools provide a fundamental understanding of these phenomena in low gravity. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | (Ed. note 1/14/2016: Compiled from reporting dated January 2016 for research performed March 2015-February 2016.) The proposed research was on gas-liquid two-phase flows through packed beds in microgravity and in support of the “Packed Bed Reactor Experiment (PBRE)” to be conducted aboard the International Space Station (ISS) during 2016. The goals of the proposed work are (i) develop hydrodynamic models that can predict the transition from bubbly flow to pulse flow for gas-liquid two-phase flow through packed-beds in microgravity (ii) extend the hydrodynamic model to examine the impact of gravity on the bubble to pulse transition (iii) use the hydrodynamic model to examine the pulse characteristics and the impact of gravity on pulse flow (iv) develop models for predicting the gas-liquid interfacial area and mass transfer rates between gas and liquid and solid and liquid and validate these by ground based experiments (both 1-g and aircraft) (v) analyze the data obtained from the ISS experiments using the hydrodynamic models developed and validate and update the models for predicting pressure drop and flow characteristics of gas-liquid two-phase flows through packed beds in microgravity with special attention to the practical region of low gas and liquid flow rates.<br><br>Progress on Stated Objectives:<br><br>(a) The results based on the first two proposed objectives are summarized in the following refereed publication, reported previously:<br><br>(1) P. Salgi and V. Balakotaiah, "Impact of Gravity on the Bubble-to-Pulse Transition in Packed Beds", AIChE Journal, 60, pp. 778-793 (2014).<br><br>(b) The Principal Investigator (PI) and post-doctoral fellow Dr. Salgi worked with the PBRE team members (Drs. B. J. Motil and E. Rame) to analyze the aircraft based two-phase flow experiments that were conducted using the engineering test rig of the PBRE. Pulse properties were analyzed and the results were summarized in the following refereed publication:<br><br>(2) P. Salgi, V. Balakotaiah, E. Rame and B. J. Motil, “Pulse Properties in Gas-Liquid Two-Phase Flow through Randomly Packed Beds Under Microgravity Conditions”, Int. J. Multiphase Flow, 73, pp. 11-16 (2015).<br><br>(c) A manuscript (P. Salgi and V. Balakotaiah, "Experimentally-Based Constitutive Relations for Co-Current Gas-Liquid Flow in Randomly Packed Beds.") based on the work done during the second year of the grant period is under preparation and will be submitted to a refereed journal within the next two months.<br><br>In addition, a presentation was made at the American Society for Gravitational and Space Research Annual Meeting, Alexandria, VA, November, 2015: P. Salgi, V. Balakotaiah, “Experimentally based Constitutive Relations for Co-current Gas-Liquid Flow in Randomly Packed Beds.” |
| **Bibliography Type:** | Description: (Last Updated: 04/26/2024) |

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| **Articles in Peer-reviewed Journals** | Salgi P, Balakotaiah V, Rame E, Motil BJ. "Pulse properties in gas–liquid flow through randomly packed beds under microgravity conditions." International Journal of Multiphase Flow. 2015 Jul;73:11-6. http://dx.doi.org/10.1016/j.ijmultiphaseflow.2015.02.020 , Jul-2015 |